

Lancaster University College
at Beijing Jiaotong University

Digital Signal Processing

Zhou Hang

Time domain analysis of discrete signals



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- Why Introduce Basic Signals and Basic Operations
- Basic Discrete Signals
- Basic Operations on Sequences

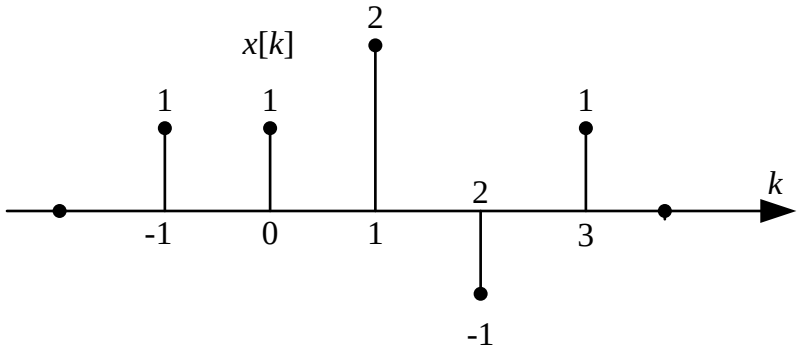
Why introduce basic signals and basic operations?

By using basic signals and basic operations, it is possible to represent complex signals, thereby transforming the analysis of complex signals into the analysis of basic signals. This is the fundamental idea of signal analysis and processing.

Basic signals and basic operations are also the fundamental carriers of signal frequency domain analysis and complex frequency domain analysis, helping us intuitively understand the correspondence and characteristics between signal time domain and transform domain.

Discrete signal representation

Figure



Vector

$$x[k]=\{1,\underset{\downarrow}{1},2,-1,1\}$$

$$x[k]=\{1,1,2,-1,1;k=-1,0,2,3\}$$

Expression

$$x[k]=2^k u[k]$$

Discrete Signal Generation

(1) Sampling of a Continuous Signal $x[k] = x(kT)$， T *is* Sampling Interval

(2) Signal as a Discrete Sequence

(3) Computer-Generated Sequence

Discrete Signal: A signal that is discrete in time (independent variable)

Digital Signal: Quantify discrete signals in terms of amplitude (function value)

Why introduce basic signals and basic operations?

◎ Basic Signal

- Pulse sequence
- Step sequence
- Index sequence
- Imaginary finger sequence
- Sine sequence
- Rectangular sequence

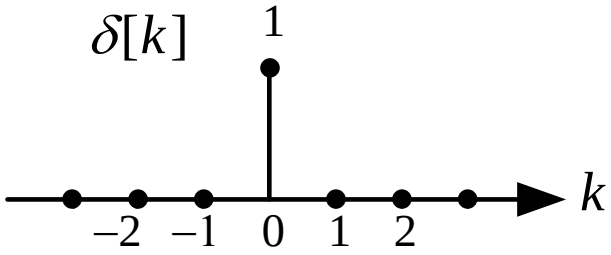
◎ Basic operations

- Reversal
- Shift
- Decimation
- Interpolation
- Convolution
- Correlation

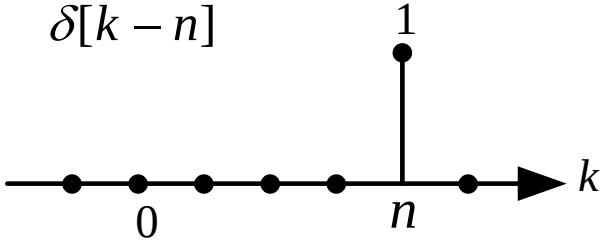
Basic discrete signal

- Unit pulse sequence

$$\delta[k]=\begin{cases}1 & k=0\\ 0 & k\neq 0\end{cases}$$

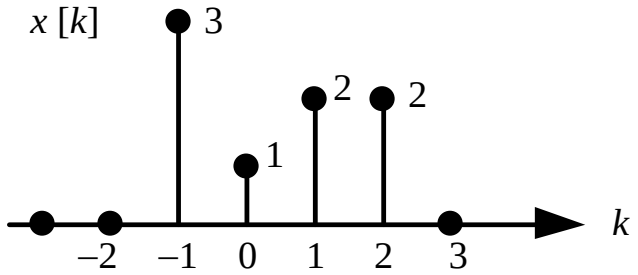


$$\delta[k-n]=\begin{cases}1 & k=n\\ 0 & k\neq n\end{cases}$$



Basic discrete signal

Function: To represent arbitrary discrete time series

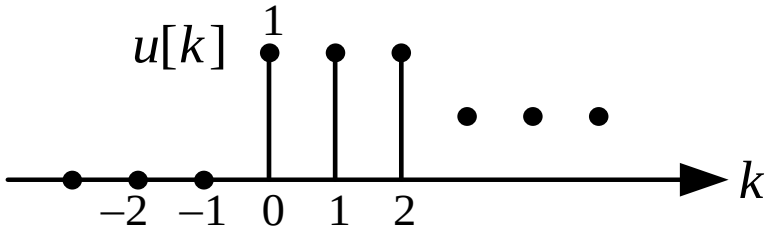


$$x[k] = 3\delta[k+1] + \delta[k] + 2\delta[k-1] + 2\delta[k-2]$$

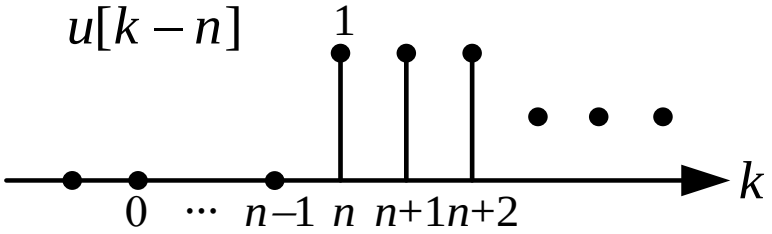
Basic discrete signal

◆ Unit Step Sequence

$$u[k]=\begin{cases}1 & k\geq 0\\ 0 & k<0\end{cases}$$



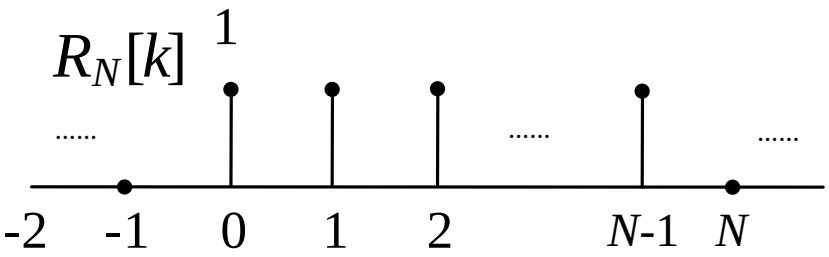
$$u[k-n]=\begin{cases}1 & k\geq n\\ 0 & k<n\end{cases}$$



Basic discrete signal

- Rectangular sequence

$$R_N[k]=\begin{cases}1 & 0\le k\le N-1\\ 0 & \text{others}\end{cases}$$



Basic discrete signal

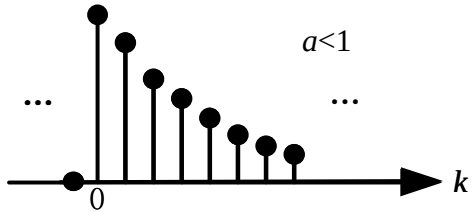
◆ Real Exponential Sequence: $x[k]=a^k, \quad k\in \mathbf{z}$

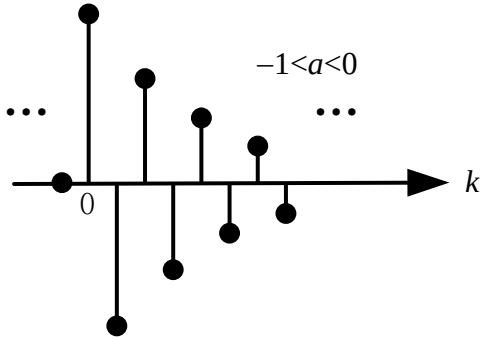
Bounded Sequence:

If $\forall k\in\mathbf{z}$, there exists$|x[k]|\le M_x$ (M_x is a constant independent of k)

$a^k u[k]$: The condition for a right-sided exponential sequence to be bounded $|a|\le 1$

$a^k u[k]$





Basic discrete signal

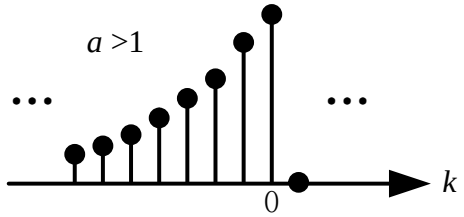
◆ Real Exponential Sequence $x[k]=a^k, \quad k \in \mathbf{z}$

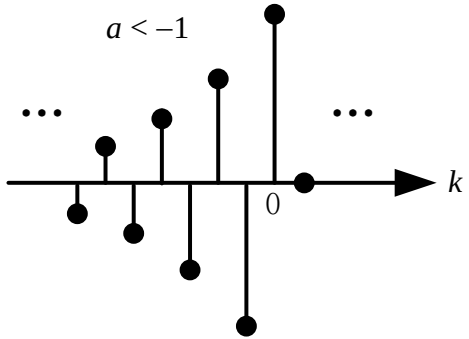
Bounded Sequence:

If $\forall k \in \mathbf{z}$, there exists $|x[k]| \le M_x$ (M_x is a constant independent of k)

$a^k u[-k]$: The condition for a left-sided exponential sequence to be bounded $|a| \ge 1$

$a^k u[-k]$





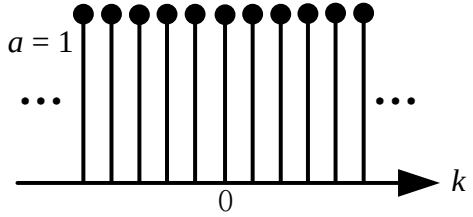
◆ Real Exponential Sequence $x[k]=a^k, \quad k \in \mathbf{z}$

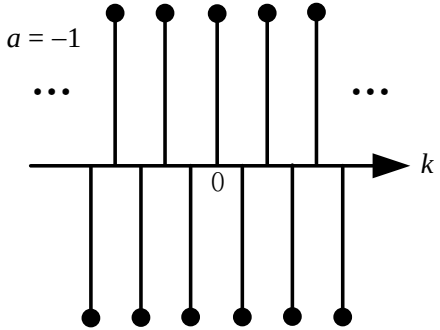
Bounded Sequence:

If $\forall k \in \mathbf{z}$, there exists $|x[k]| \le M_x$ (M_x is a constant independent of k)

a^k: The condition for the boundedness of bilateral index sequences $|a| = 1$

a^k





◆ Virtual exponential sequence and sine sequence

$$x[k]=\mathrm{e}^{\mathrm{j}\Omega_0 k} \qquad x[k]=A\cos(\Omega_0 k+\phi)$$

The Euler formula can be used to link the imaginary exponential sequence with the sine sequence, that is

$$\mathrm{e}^{\mathrm{j}\Omega_0 k}=\cos\Omega_0 k+\mathrm{j}\sin\Omega_0 k$$

$$\cos\Omega_0 k=\frac{1}{2}(\mathrm{e}^{\mathrm{j}\Omega_0 k}+\mathrm{e}^{-\mathrm{j}\Omega_0 k})$$

$$\sin\Omega_0 k=\frac{1}{2\mathrm{j}}(\mathrm{e}^{\mathrm{j}\Omega_0 k}-\mathrm{e}^{-\mathrm{j}\Omega_0 k})$$

Basic discrete signal

$e^{j\Omega_0 k}$ can be obtain by $e^{j\omega_0 t}$ obtained through sampling

$$e^{j\Omega_0 k} = e^{j\omega_0 t}\Big|_{t=kT}, \quad \Omega_0 = \omega_0 T$$

Periodic:

If $e^{j\Omega_0(k+N)} = e^{j\Omega_0 k} e^{j\Omega_0 N} = e^{j\Omega_0 k}$, there $e^{j\Omega_0 N} = 1$

also $\Omega_0 N = M2\pi$, M is an integer, The signal is a periodic signal。

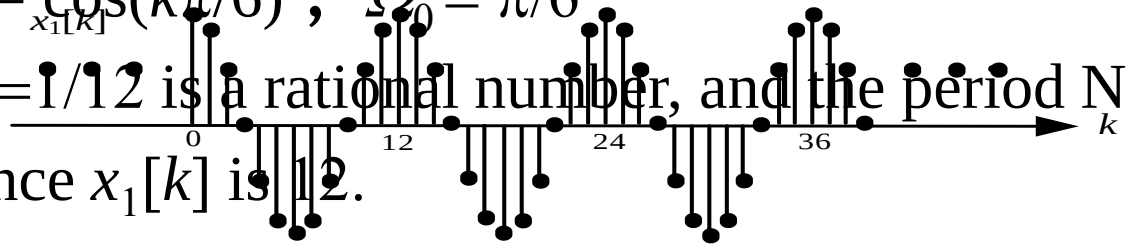
If $\Omega_0/2\pi = M/N$, the discrete signal is a periodic sequence;

when N、M is an irreducible integer, The period of the signal is N。

Try to analyze whether the following sinusoidal signals are periodic signals.

Solution: (1) For $x(t)=\cos(t)$, the sequence $x_1[k]$ obtained by sampling with T=p/6 (s)

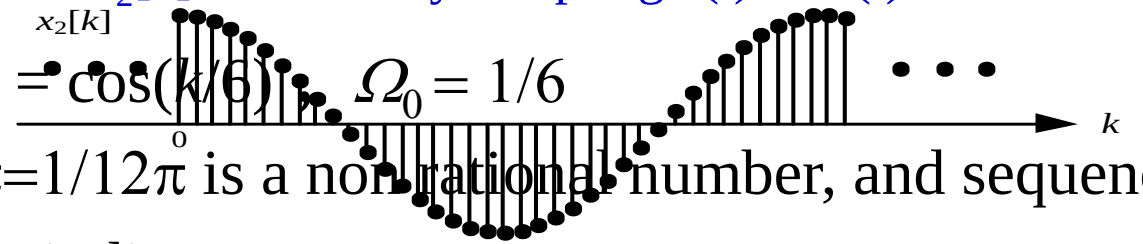
$x_1[k]=\cos(k\pi/6)$, $\Omega_0=\pi/6$



$\Omega_0/2\pi=1/12$ is a rational number, and the period N of sequence $x_1[k]$ is 12.

The sequence $x_2[k]$ obtained by sampling $x(t)=\cos(t)$ with T=1/6 (s)

$x_2[k]=\cos(k/6)$, $\Omega_0=1/6$



$\Omega_0/2\pi=1/12\pi$ is a non-rational number, and sequence $x_2[k]$ is a non periodic sequence.

Discrete sine signals or imaginary exponential signals are not necessarily periodic signals.

Try to analyze whether the following sinusoidal signals are periodic signals.

Solution: (1) The sequence $x_1[k]=\cos(0.08\pi k)$ obtained by sampling the continuous signal $\cos(\pi t)$ at $T=0.08$ (s)

$$\frac{\Omega_0}{2\pi}=\frac{0.08\pi}{2\pi}=\frac{1}{25}=\frac{M}{N}$$ $M=1$

The period of sequence $x_1[k]$ is $N=25$

(2) The sequence $x_2[k]=\cos(0.16\pi k)$ obtained by sampling the continuous signal $\cos(\pi t)$ at $T=0.16$ (s)

$$\frac{\Omega_0}{2\pi}=\frac{0.16\pi}{2\pi}=\frac{2}{25}=\frac{M}{N}$$ $M=2$

The period of sequence $x_2[k]$ is $N=25$

(3) The sequence $x_3[k]=\cos(0.24\pi k)$ obtained by sampling the continuous signal $\cos(\pi t)$ at $T=0.24$ (s)

$$\frac{\Omega_0}{2\pi}=\frac{0.24\pi}{2\pi}=\frac{3}{25}=\frac{M}{N}$$ $M=3$

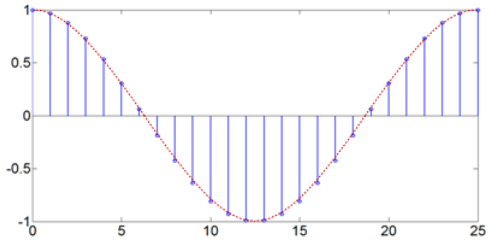
The period of sequence $x_3[k]$ $N=25$

What is the meaning of the value of M?

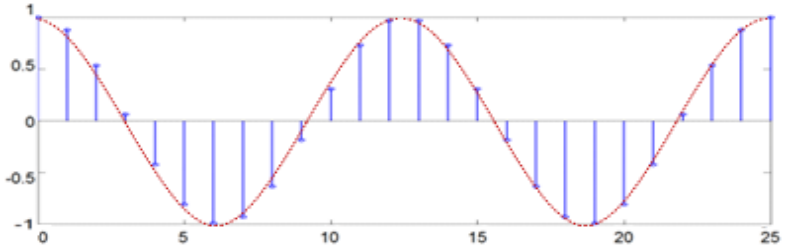
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$\frac{M}{N}=\frac{1}{25}$



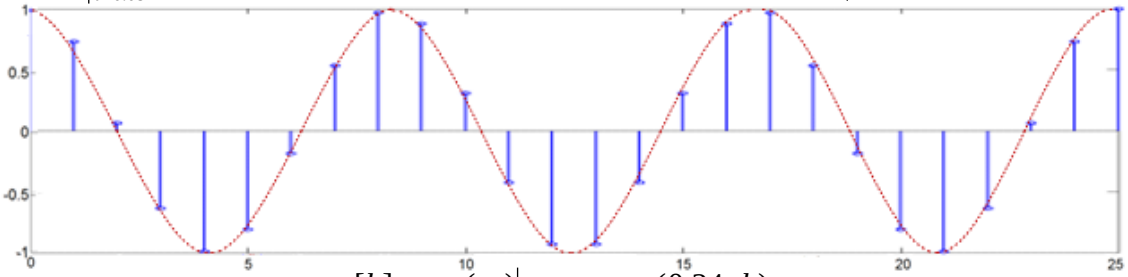
$x_1[k]=\cos(\pi t)\big|_{t=0.08k}=\cos(0.08\pi k)$



$\frac{M}{N}=\frac{2}{25}$

$x_2[k]=\cos(\pi t)\big|_{t=0.16k}=\cos(0.16\pi k)$

$\frac{M}{N}=\frac{3}{25}$



$x_3[k]=\cos(\pi t)\big|_{t=0.24k}=\cos(0.24\pi k)$

Conclusion: When $\Omega_0/2\pi=M/N$、 N and M are irreducible integers, then N is the period of the discrete sequence, and M represents the number of periods of the original continuous periodic signal contained within one period N of the discrete sequence.

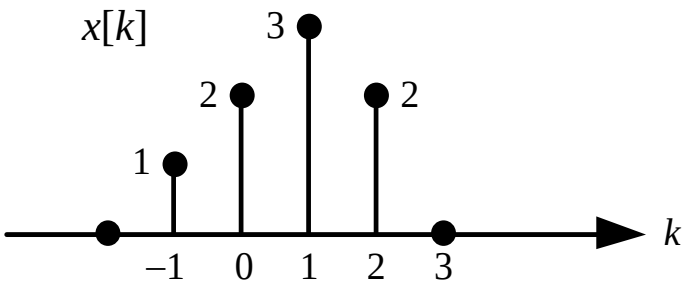
Basic operations of sequences

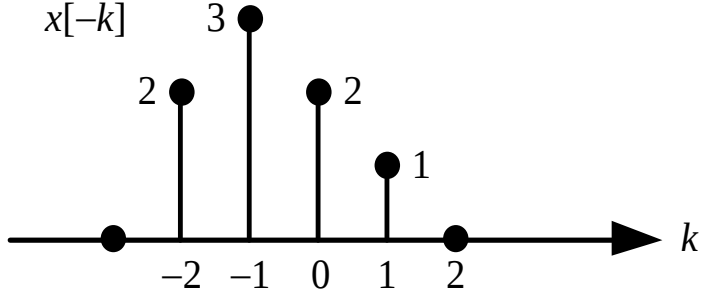
- Reversal
- Shift
- Decimation
- Interpolation
- Convolution
- Correlation

Basic operations of sequences

➢Time Reversal $x[k] \rightarrow x[-k]$

Perform a 180° rotation of $x[k]$ around the vertical axis.





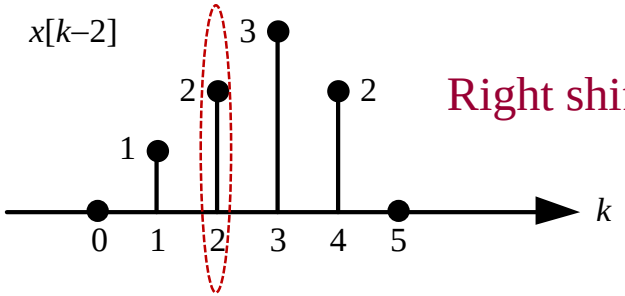
Basic operations of sequences

➢ Shift

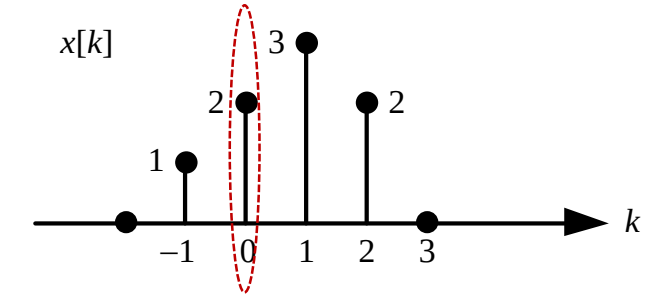
$$x[k] \rightarrow x[k \pm n]$$

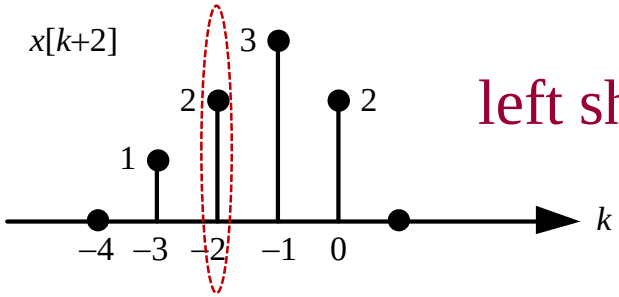
$x[k-n]$ represents shifting $x[k]$ to the right by n units in signal processing.

$x[k+n]$ represents shifting $x[k]$ to the left by n units in signal processing.



Right shift





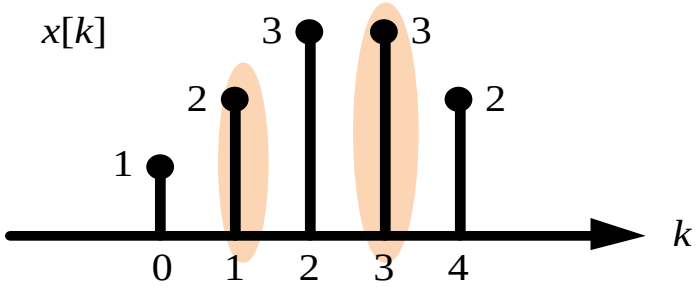
left shift

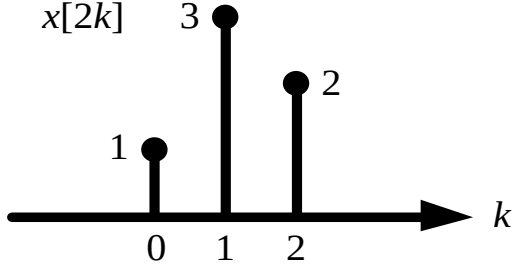
Basic operations of sequences

➢ Decimation

$x_D[k] \rightarrow x[Mk]$ M is a positive integer

Decimate one point every M points (M–1 points) in the original sequence





Basic operations of sequences

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Original signal x

```
[x,Fs,bits] = wavread('天路');
% Fs=44,100 Hz
```



After 4-fold decimation, the signal is denoted as x_1.

```
x1 = downsample(x, 4);
% Fs=44,100/4 Hz
```



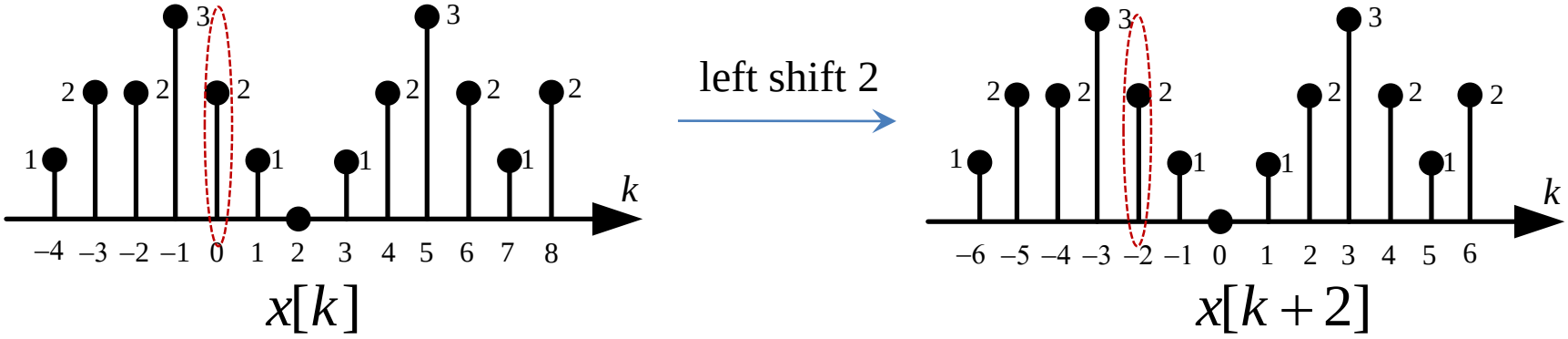
After 8-fold decimation, the signal is denoted as x_2.

```
x2 = downsample(x, 8);
% Fs=44,100/8 Hz
```

Basic operations of sequences

E.g The discrete signal $x[k]$ as illustrated in the figure, please describe or sketch the graph of $x[-3k+2]$.

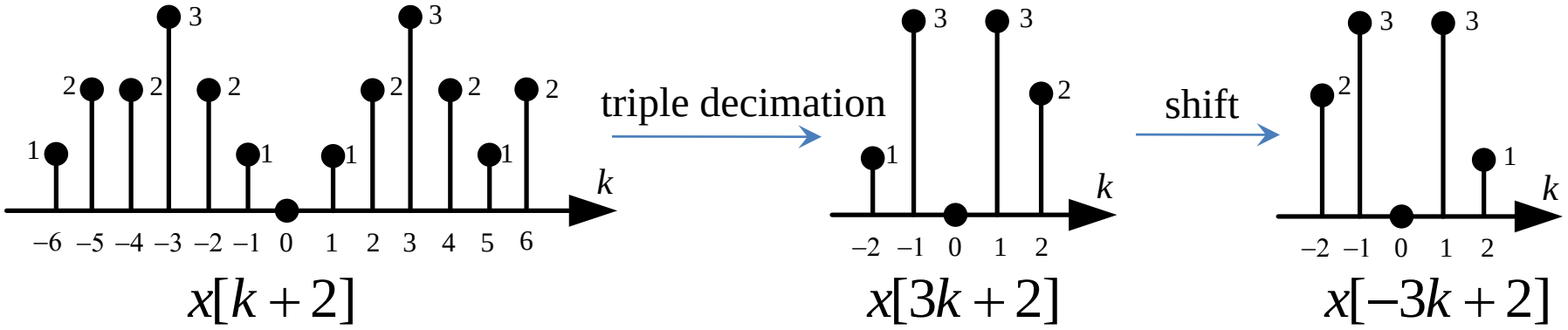
Solution : The expression $x[-3k+2]$ involves operations of shifting, decimation, and flipping, can first decimate x $x[-3k+2]$ by tripling, and then shift it to obtain $x[-3k+2]$.



Basic operations of sequences

Given the discrete signal $x[k]$ as shown in the figure, draw the graph of $x[-3k+2]$

Solution:

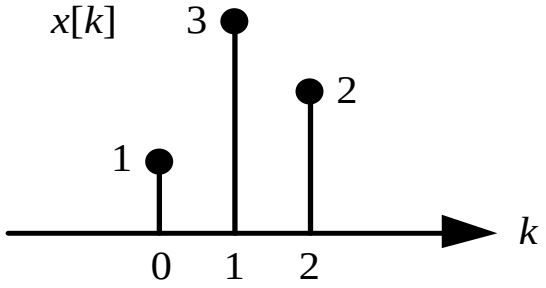


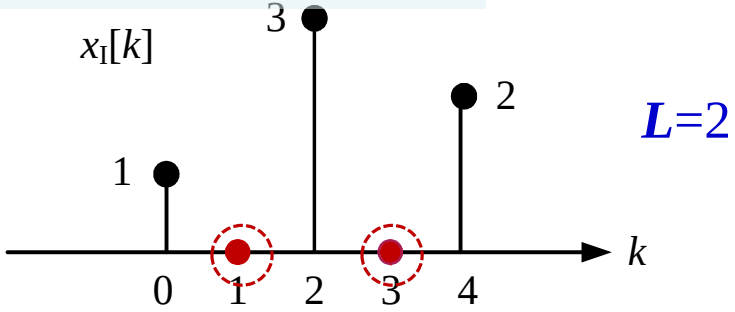
Basic operations of sequences

➢ Interpolation

$$x_{\mathrm{I}}[k]=\begin{cases} x[k/L] & k\text{是}L\text{的整数倍} \\ 0 & \text{其它} \end{cases}$$

Insert $L-1$ points between each point in the original sequence



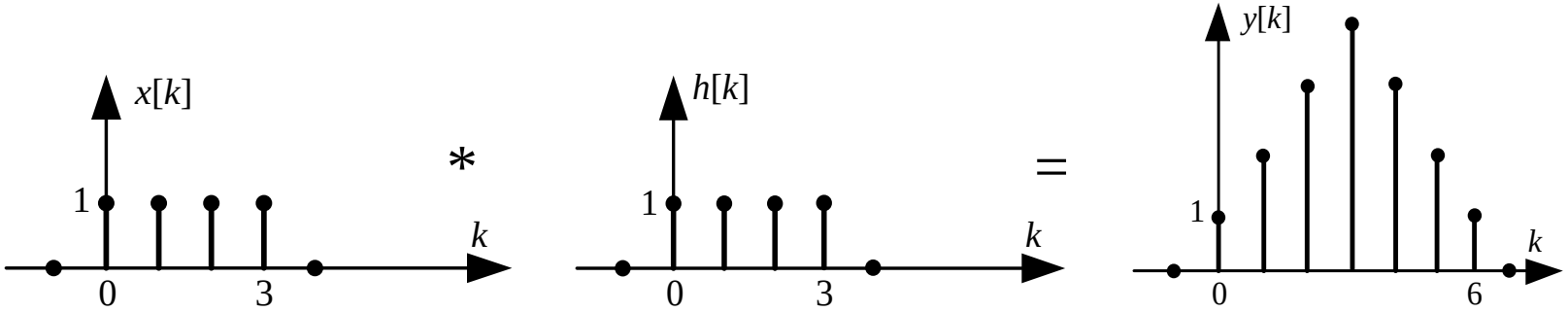


***L*=2**

Basic operations of sequences

➢ Convolution $$y[k]=\sum_{n=-\infty}^{\infty}x[n]h[k-n]$$

Convolution calculation mainly includes graphical method, analytical method, and list method, which deepen the understanding of the interaction between signals and systems through convolution calculation, rather than calculating for the sake of calculation.



Basic operations of sequences

[E.g] Calculate $x[k]=\alpha^k u[k]$ with $h[k]=\beta^k u[k]$ convolution。

Solution: $x[k]*h[k]=\alpha^k u[k]*\beta^k u[k]$

$$y[k]=\sum_{n=-\infty}^{\infty} x[n]h[k-n]$$

$$=\sum_{n=-\infty}^{\infty}\alpha^n u[n]\cdot\beta^{k-n}u[k-n]$$

$$=\begin{cases}\sum_{n=0}^{k}\alpha^n\cdot\beta^{k-n}, k\geq 0\\ 0, \quad k<0\end{cases}=\begin{cases}\dfrac{\beta^{k+1}-\alpha^{k+1}}{\beta-\alpha}u[k], \alpha\neq\beta\\ (k+1)\alpha^k u[k], \ \alpha=\beta\end{cases}$$

Basic operations of sequences

[E.g] Calculate $x[k]=\{1,2,\underset{\downarrow}{0},3,2\}$ with $h[k]=\{1,\underset{\downarrow}{4},2,3\}$ convolution

Solution:

		$x[-2]$	$x[-1]$	$x[0]$	$x[1]$	$x[2]$
		1	2	0	3	2
$h[-1]$	1	1	2	0	3	2
$h[0]$	4	4	8	0	12	8
$h[1]$	2	2	4	0	6	4
$h[2]$	3	3	6	0	9	6

	$x[k]$	1	2	0	3	2		
$*$	$h[k]$		1	4	2	3		
			3	6	0	9	6	
		2	4	0	6	4		
	4	8	0	12	8			
1	2	0	3	2				
1	6	10	10	20	14	13	6	

$$y[k]=\{1,6,10,\underset{\downarrow}{10},20,14,13,6\}$$

Definition of correlation :

Interrelated: $R_{xy}[n]=\sum_{k=-\infty}^{\infty}x[k]y[k+n]$

Autocorrelation: $R_{x}[n]=\sum_{k=-\infty}^{\infty}x[k]x[k+n]$

Related operations have been widely applied in signal detection and spectral analysis

The relationship between signal correlation operation and signal convolution operation :

$$R_{xy}[n] = \sum_{k=-\infty}^{\infty} x[k]y[k+n]$$

$$= \sum_{k=-\infty}^{\infty} x[k-n]y[k]$$

$$= \sum_{k=-\infty}^{\infty} x[-(n-k)]y[k]$$

$$= x[-n] * y[n]$$

$$x[k] * y[k] = \sum_{n=-\infty}^{\infty} x[k-n]y[n]$$

[E.g] Calculate the cross-correlation $R_{xy}[n]$ and $R_{yx}[n]$ between $y[k]=\{-1,2,1,-1\}$ and $x[k]=\{2,1,-2,1\}$ as well as the autocorrelation $R_x[n]$ of sequence $x[k]$.

Solution:

$$R_{xy}[n]=\sum_{k=-\infty}^{\infty}x[k]y[k+n]$$

$$=x[-n]*y[n]$$

$$=\{-1,4,-4,-3,7,1,-2\}$$

$x[-n]$			1	−2	1	2	
$y[n]$			−1	2	1	−1	
			−1	2	−1	−2	
		1	−2	1	2		
	2	−4	2	4			
−1	2	−1	−2				
−1	4	−4	−3	7	1	−2	

[E.g] Calculate the cross-correlation $R_{xy}[n]$ and $R_{yx}[n]$ between $y[k]=\{-1,2,\underset{\downarrow}{1},-1\}$ and $x[k]=\{2,\underset{\downarrow}{1},-2,1\}$ as well as the autocorrelation $R_x[n]$ of sequence $x[k]$.

Solution: Similarly, $R_{yx}[n]$ can be solved

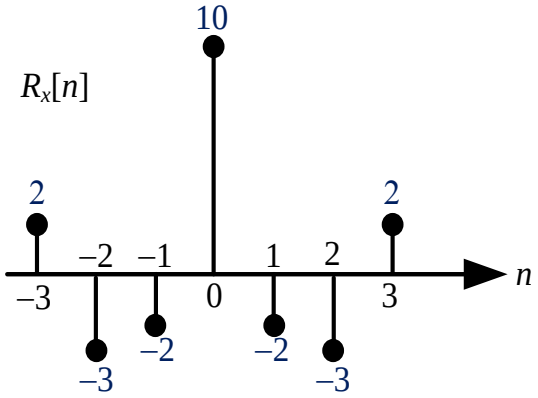
$$R_{yx}[n]=\sum_{k=-\infty}^{\infty} y[k]x[k+n]=y[-n]*x[n]=\{-2,1,7,\underset{\downarrow}{-3},-4,4,-1\}$$

$$R_{xy}[n]=\sum_{k=-\infty}^{\infty} x[k-n]y[k]=x[-n]*y[n]=\{-1,4,-4,\underset{\downarrow}{-3},7,1,-2\}$$

$$R_{xy}[n]=R_{yx}[-n]$$

[E.g] Calculate the cross-correlation $R_{xy}[n]$ and $R_{yx}[n]$ between $x[k]=\{\underset{\downarrow}{2},1,-2,1\}$ and $y[k]=\{\underset{\downarrow}{-1},2,1,-1\}$, as well as the autocorrelation $R_x[n]$ of sequence $x[k]$.

Solution: $R_x[n]=\sum_{k=-\infty}^{\infty}x[k]x[k+n]=x[-n]*x[n]=\{2,-3,-2,\underset{\downarrow}{10},-2,-3,2\}$



$$R_x[n]=R_x[-n]$$

$$R_x[0]\geq|R_x[n]|$$

Characteristics of signal related operations

$$R_{xy}[n] = x[-n] * y[n]$$

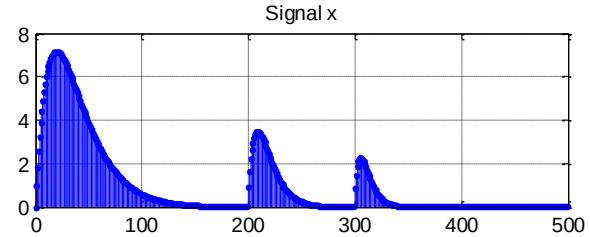
$$R_{xy}[n] = R_{yx}[-n]$$

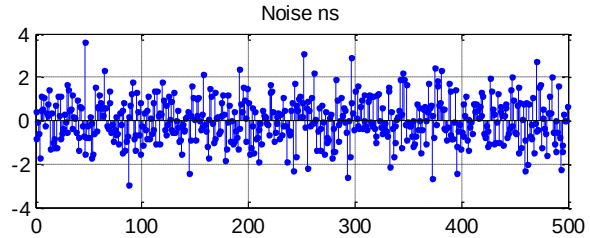
$$R_x[n] = R_x[-n]$$

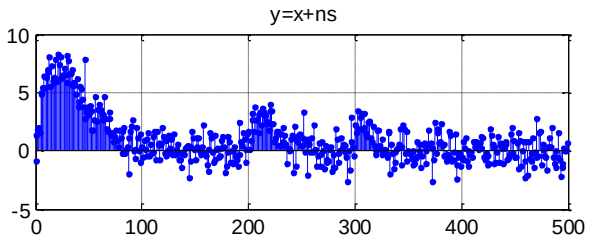
$$R_x[0] \geq |R_x[n]|$$

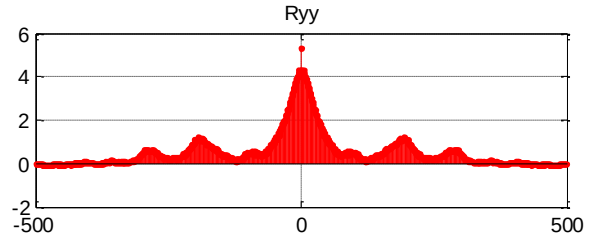
Sequence related operations - signal detection

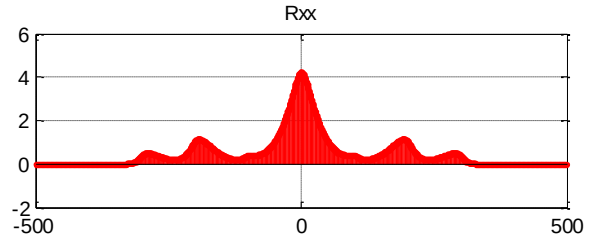
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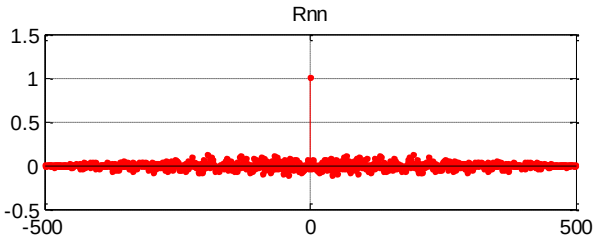




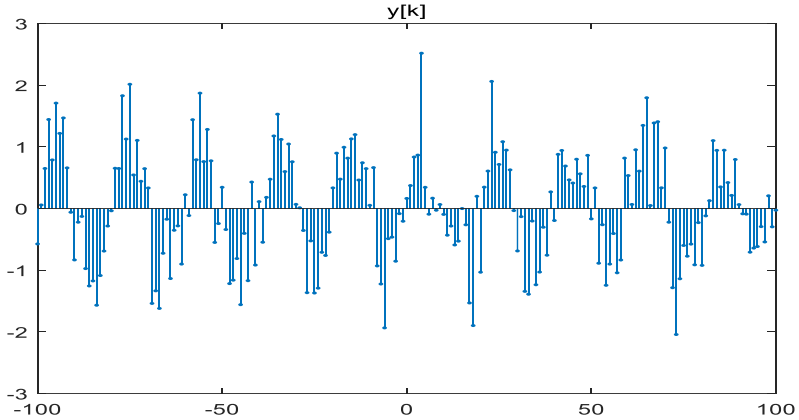


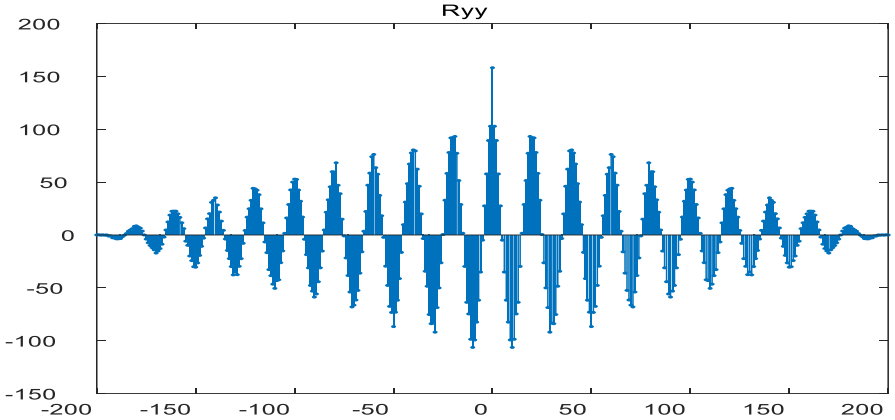


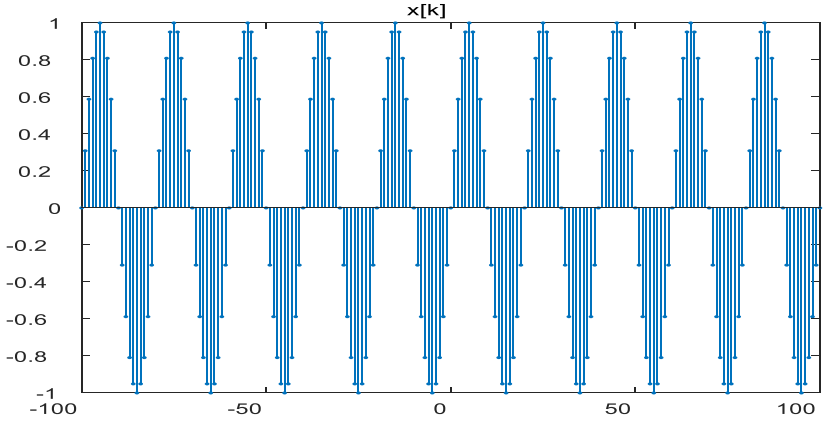


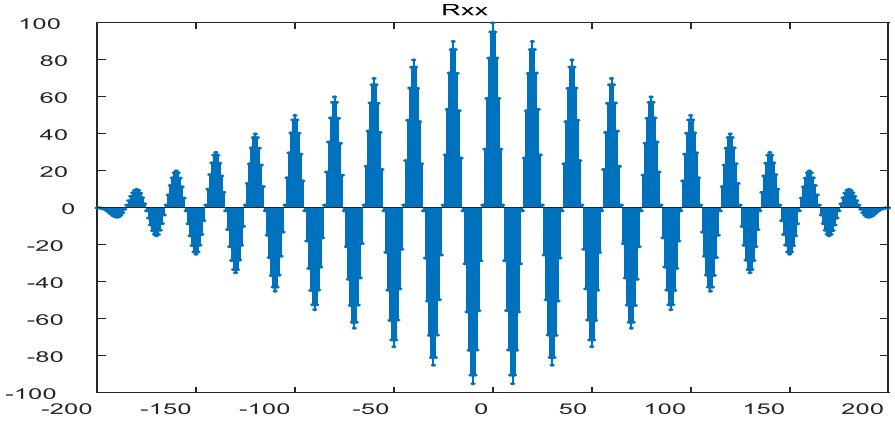


Sequence related operations - signal detection



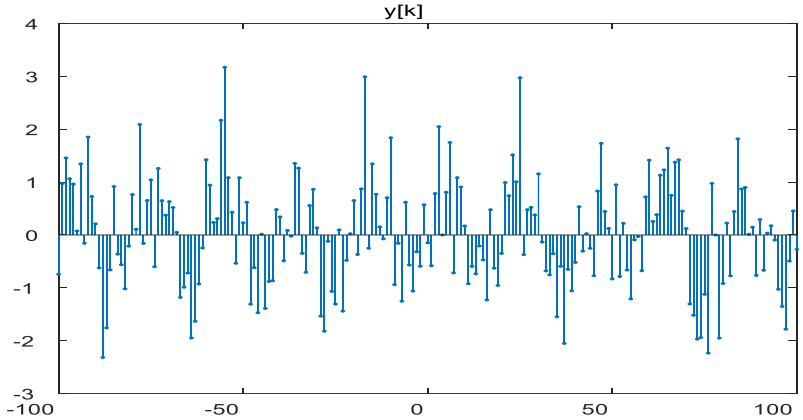


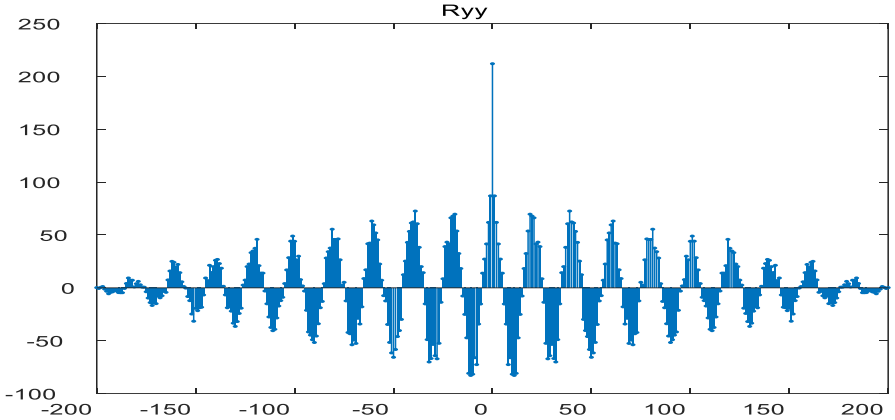


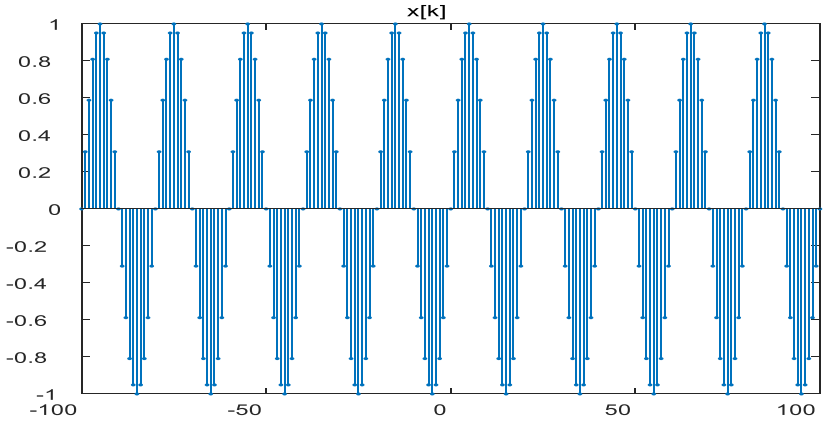


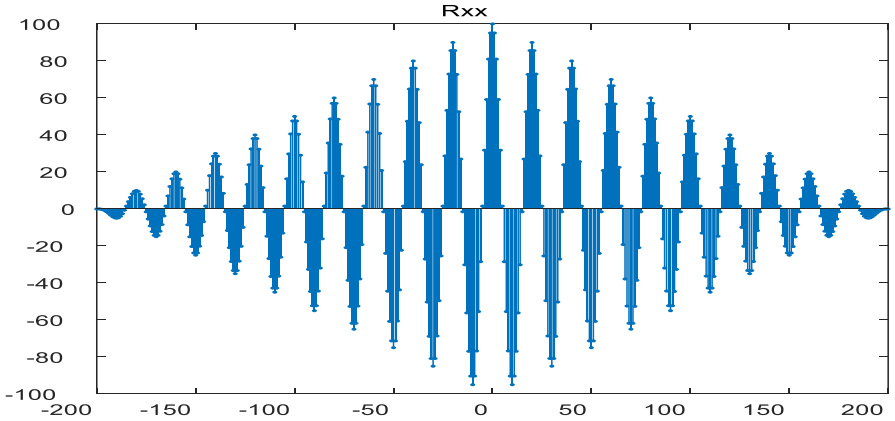
Sequence related operations - signal detection

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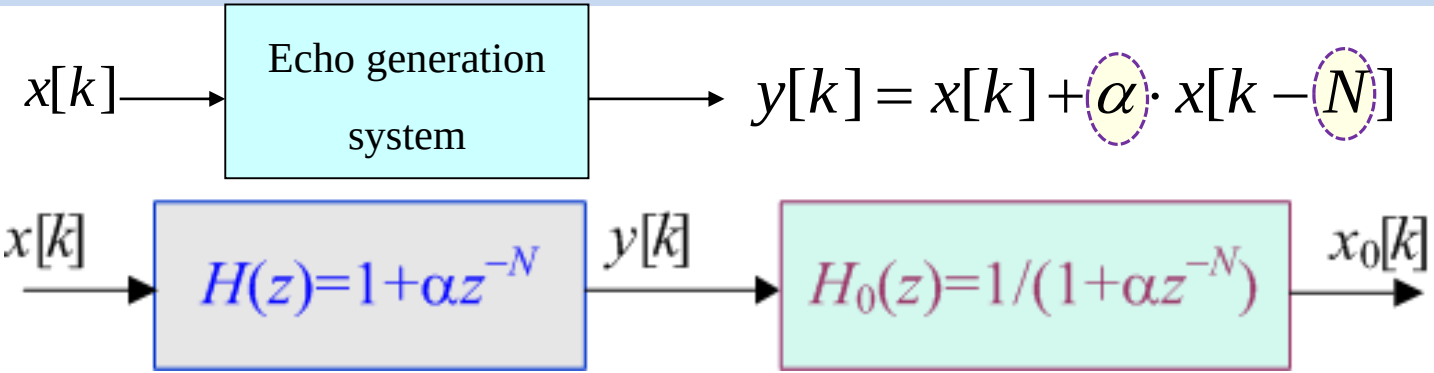








Sequence related operations - echo cancellation



$$y[k] = x[k] + \alpha \cdot x[k-N]$$

$$H(z) = 1 + \alpha\, z^{-N}$$

$$x_0[k] + \alpha \cdot x_0[k-N] = y[k]$$

$$H_0(z) = 1/H(z) = 1/(1 + \alpha\, z^{-N})$$

$$x_0[k] = y[k] - \alpha \cdot x_0[k-N]$$

Sequence related operations - echo cancellation


```
load mtlb; x = mtlb;
M = length(x);
a = 0.9; N = 3000;
x0 = [x', zeros(1, N)];
xN = [zeros(1, N), x'];
y = x0 + a.*xN;  % Generate Echo
subplot(2, 2, 1); stem(x0, '.');
title('Signal x[k])'); grid on
subplot(2,2,2);  stem(y,'.');
title('y[k]=x[k] + 0.9*x[k-N]');  grid on
```

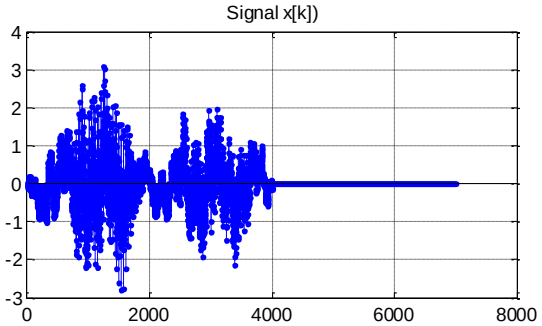
```
[Rxx,n] = xcorr(x0,x0);
subplot(2, 2, 3);
stem(n,Rxx,'r.');  title('Rxx[n]');
grid on

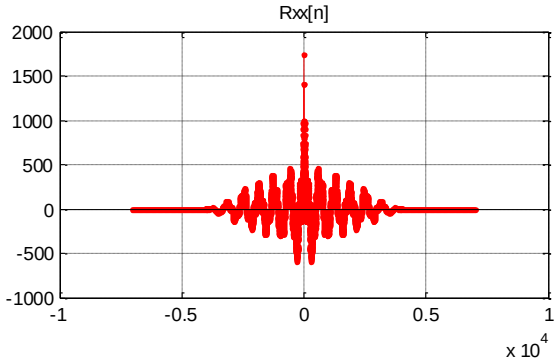
[Ryy,n] = xcorr(y,y);
subplot(2, 2, 4);
stem(n,Ryy,'r.');  title('Ryy[n]');
 grid on
```

Sequence related operations - echo cancellation

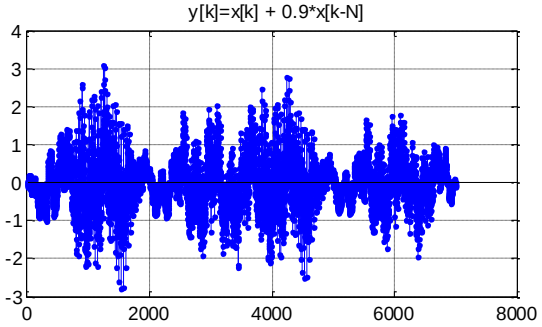
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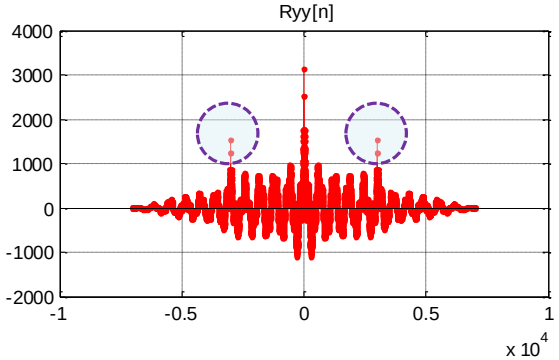










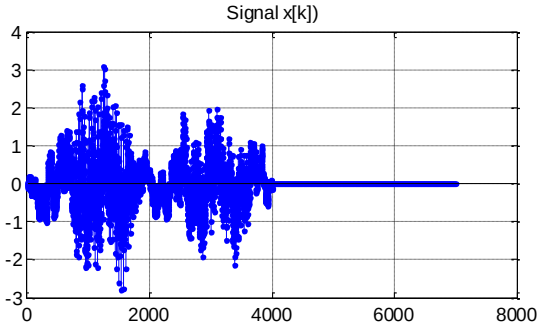


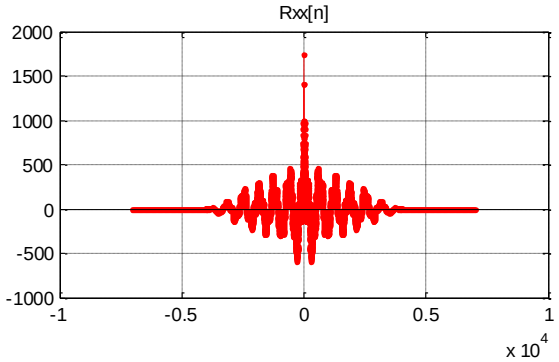
$$y[k] = x[k] + \alpha \cdot x[k - N]$$

Sequence related operations - echo cancellation

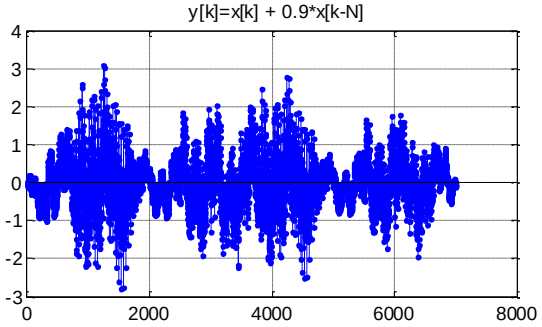
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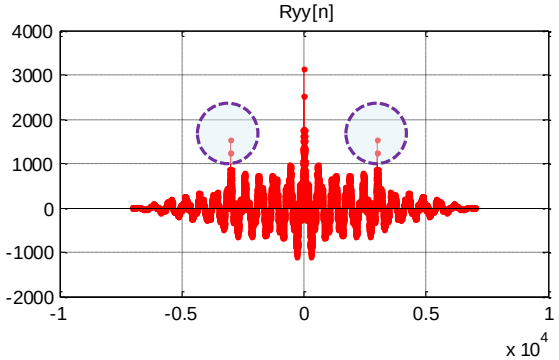






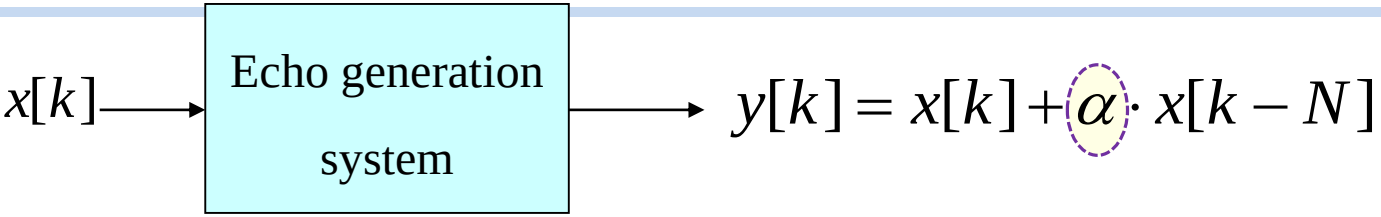






$$y[k] = x[k] + \alpha \cdot x[k - N]$$

Sequence related operations - echo cancellation



$$R_y[n] = \sum_{k=-\infty}^{\infty} y[k]y[k+n]$$

$$= \sum_{k=-\infty}^{\infty} (x[k] + \alpha x[k-N])(x[k+n] + \alpha x[k-N+n])$$

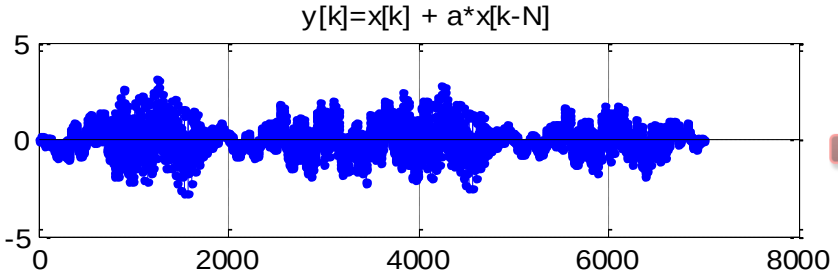
$$= (1+\alpha^2)R_x[n] + \alpha R_x[n-N] + \alpha R_x[n+N]$$

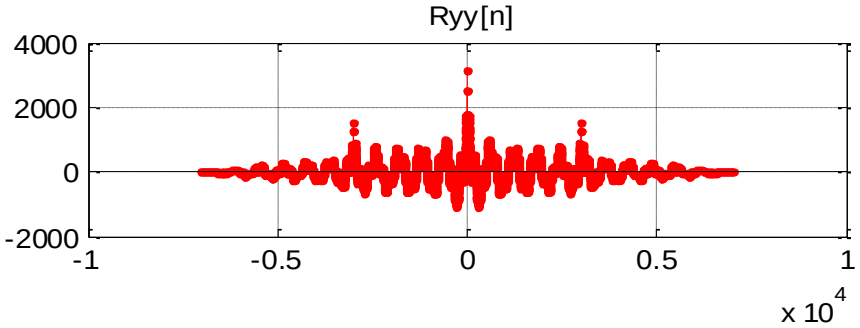
Through deduction, it can be concluded that:

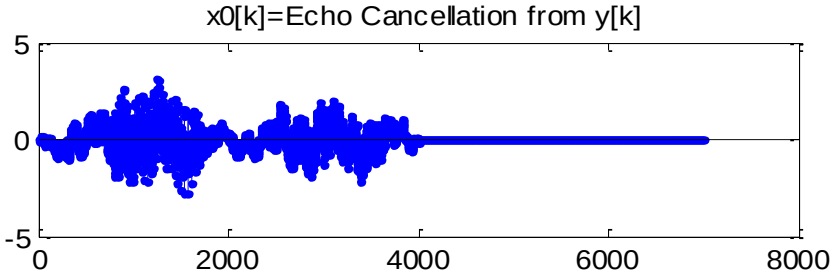
$$R_y[0] / R_y[N] \approx (1+\alpha^2) / \alpha$$

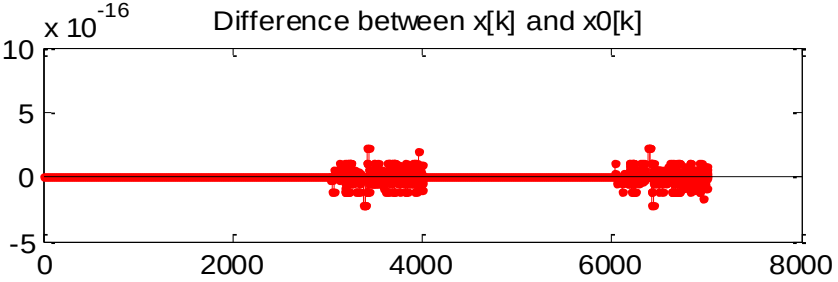
Sequence related operations - echo cancellation

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Thanks!

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude!